RAK19026 VC WisMesh Base Board Datasheet

IMPORTANT

This documentation is for board version VC. For information on other board versions, check the referring datasheet:

- Board Version VA

Overview

RAK19026 VC is a **WisBlock Base Board** that connects **WisBlock IO**, and **WisBlock Modules**. It provides the power supply and interconnection to the modules attached to it.
Different to other WisBlock Base Boards, it has no Core Slot for a WisBlock Core module. Instead it has a WisDuo RAK4630 integrated as its MCU. Beside of the MCU, a GNSS module and acceleration sensor are as well on the Base Board. Similar to other WisBlock Base Boards it has two slots C-D for WisBlock modules. The WisBlock modules are attached to the top side of the RAK19026 VC. The Slot C and D support 10 mm WisBlock modules. In addtion it has one IO slot for WisBlock IO modules.

For convenience, there is a USB-C connector that is connected directly to WisDuo MCU’ s USB port. It can be used for uploading firmware or serial communication. The USB-C connector is also used as a battery charging port.

WisBlock modules are connected to the RAK19026 VC WisBlock Base board via high-speed board-to-board connectors. They provide secure and reliable interconnection to ensure the signal integrity of each data bus. A set of screws are used for fixing the modules, which makes it reliable even in an environment with lots of vibrations.

RAK19026 VC has connectors for the following:

- 1 WisBlock modules compatible with IO slot
- 2 WisBlock modules compatible with slots C-D
- 1 Type-C USB port for programming and debugging
- 3.7 V Rechargeable battery connector
  - 3 pin connector for EU/UK markets (CE & UKCA certifications)
  - 2 pin connector for other markets
- 5 V Solar panel connector
- Multiple headers with solder contacts
  - 4 pin header with SCL, SDA, AIN1, and IO8
  - 4 pin header with IO3, IO4, IO6, and IO7
  - 4 pin header with 3V3_S, GND, 3.3V, and GND
  - 4 pin header for OLED display with I2C, 3.3V, and GND
  - 4 pin header with SWD, SWCLK, RST, and GND (for programming and debugging with RAKDAP1 or Jlink adapters)

Additionally, it has two user-definable LEDs, one power supply/charging indicator LED, a reset button, a user-definable button and a battery disconnection switch.

NOTE

Battery Disconnection Switch
The battery can still be recharged through the USB port or the 5 V/Solar panel connector, even if the switch is in OFF position.

If modules require to be placed outside of the WisMesh Base Board, extension cables are available: RAK19005 WisBlock Sensor Extension Cable to position the WisBlock Sensor modules apart from the WisBlock Base board or in any part of your case. RAK19008 WisBlock IO Extension Cable to position the WisBlock IO modules apart from the WisBlock Base board or in any part of your case.

If you can't find a WisBlock module that fits your IoT requirements, use the standard connectors of WisBlock to develop your specific function module. WisBlock supports open-source hardware architecture and you can find tutorials showing how to create your own Awesome WisBlock module.

[redacted]

- [redacted]

Main Features

- Flexible building block design, which enables modular function realization and expansion
- High-speed interconnection connectors in the WisMesh Base Board to ensure signal integrity
- Features the RAK4630 WisDuo module with the Nordic nRF52840 MCU and the Semtech SX1262 LoRa transceiver
- Supports multiple types of sensors. A single board can support a combination of two different types of sensors
- Low power battery power supply
- Supports lithium battery charging
- Supports solar panel charging
- Fulfills industrial level design
- User definable button
- Battery disconnection switch
- Integrated GNSS module and acceleration sensor
- Connector for OLED display
- Compact size: 41.6 x 60 mm

Specifications

Overview

There are three (3) slots on RAK19026 VC WisMesh Base:

- **IO Slot**: This slot is used for IO extension modules.
- **Two Sensor Slots**: The sensor slots C and D are used to connect with the I2C bus, have two GPIO's each and an UART RX/TX.

Also, there are multiple 2.54 mm pitch hole pads for extension interfaces.

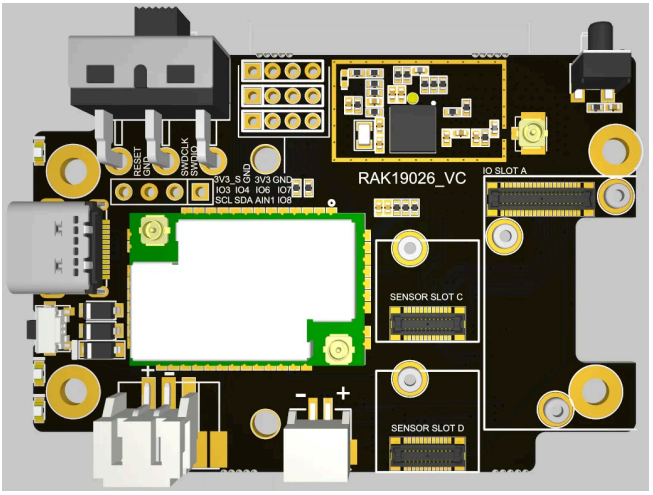


Figure 1: WisMesh Base top view

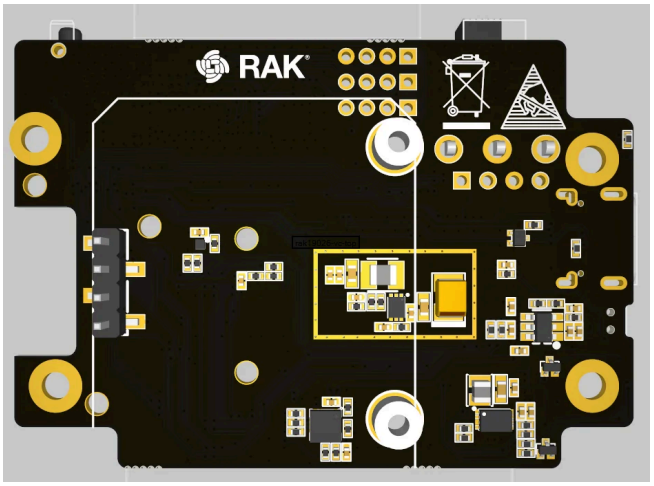


Figure 2: WisMesh Base bottom view

Block Diagram

The block diagram shows the internal architecture and external interfaces of the RAK19026 VC board.

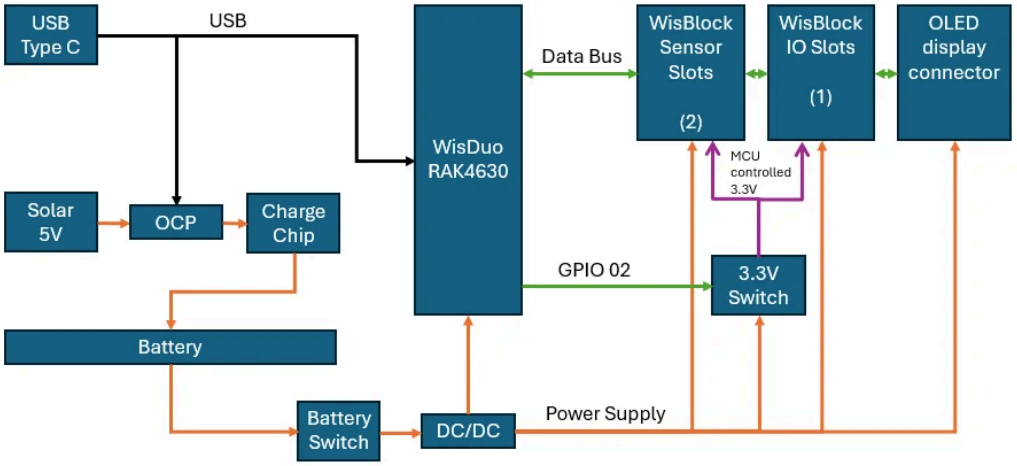


Figure 3: RAK19026 VC WisMesh Base block diagram

Hardware

The hardware specification is categorized into six parts. It shows the interfacing, pinouts and the corresponding functions and diagrams. It also presents the electrical, environmental, and mechanical parameters that include the tabular data of the functionalities and standard values of the RAK19026 WisMesh Base.

Interfaces

RAK19026 VC WisMesh Base provides the following interfaces, headers, a button, and WisBlock Connectors.

- One Type-C USB connector
- One connector for the IO slot
- Two connectors for WisBlock sensor modules (slots C and D)
- Multiple pin header 2.54 mm hole pads
  - 4 pin header with SCL, SDA, AIN1, and IO8
  - 4 pin header with IO3, IO4, IO6, and IO7
  - 4 pin header with 3V3_S, GND, 3.3V, and GND
  - 4 pin header for OLED display with I2C, 3.3V, and GND
  - 4 pin header with SWD, SWCLK, RST, and GND (for programming and debugging with RAKDAP1 or Jlink adapters)

- Battery connector
  - 3 pin connector for EU/UK markets (CE & UKCA certifications)
  - 2 pin connector for other markets
- 2-pin solar panel interface

Additionally, the RAK19026 VC has two user-definable LEDs, one power supply/charging indicator LED, a reset button, a user-definable button and a battery ON/OFF switch.

Figure 4 and **Figure 5** shows the location of RAK19026 VC main components:

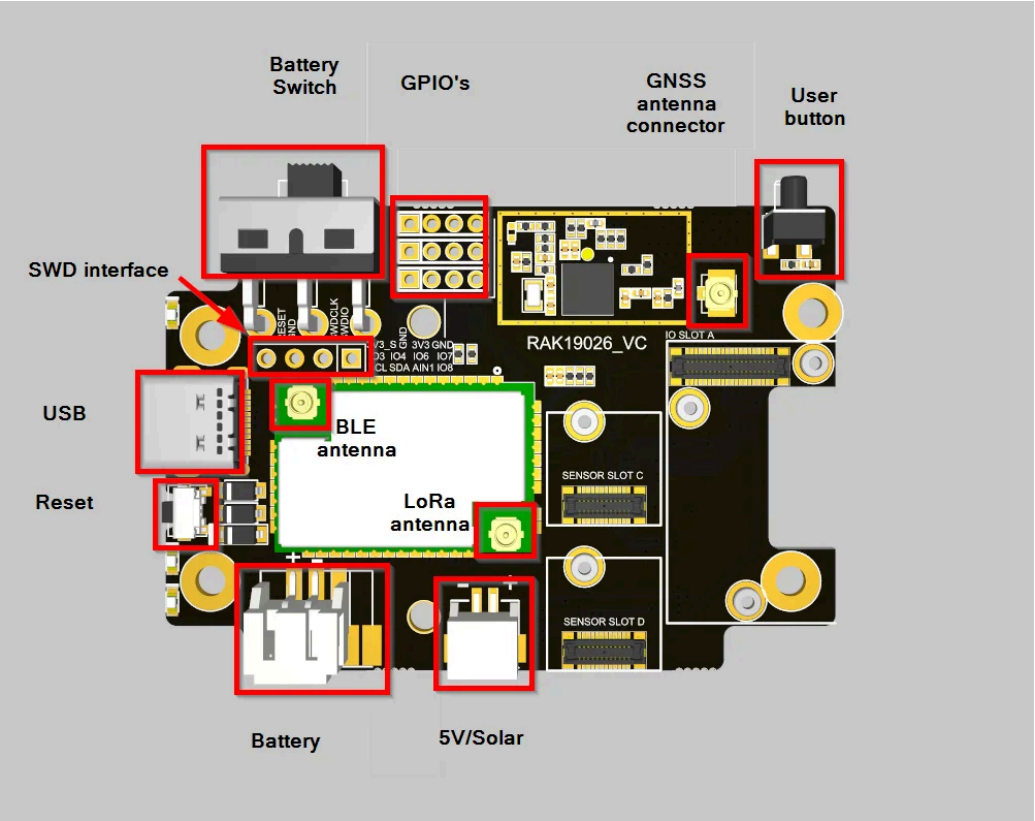


Figure 4: RAK19026 VC top view components

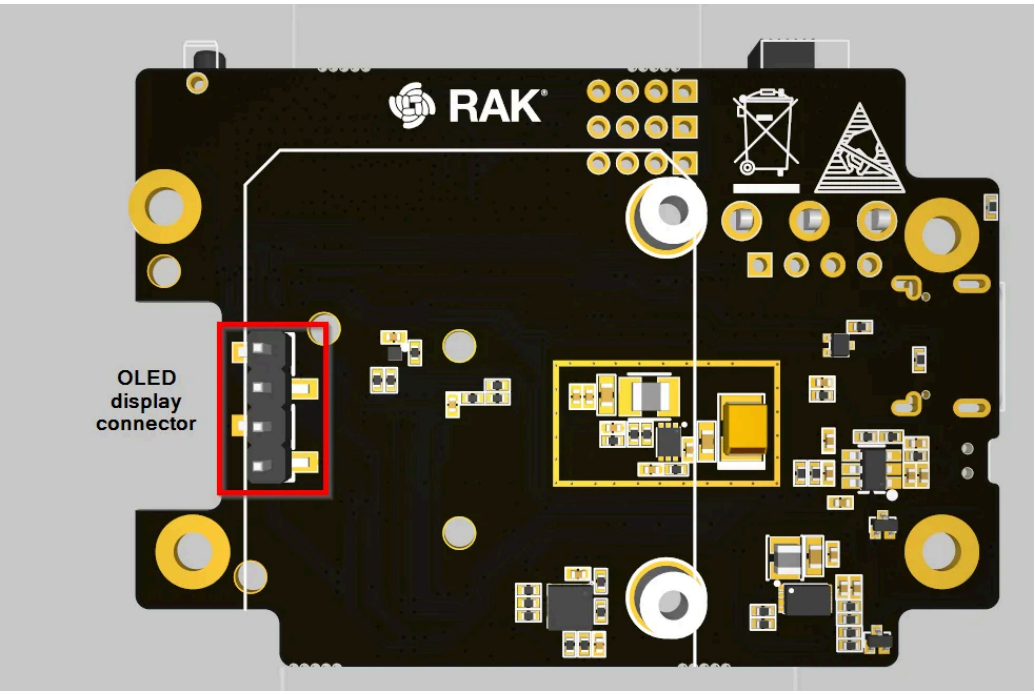


Figure 5: RAK19026 VC bottom view components

Type-C USB port

The Type-C USB connector is compliant with the USB 2.0 specification. This USB interface directly communicates with the connected **WisBlock Core** module. It is also used as a charging input port for the battery. Here are some of the advantages of the Type-C USB connector:

- Smaller and reversible connector shape
- Port can be input or output
- Fast battery charging

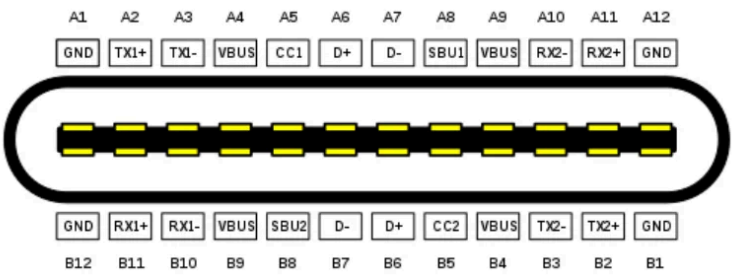


Figure 6: USB Type-C receptacle pinout

Pin Headers

On the RAK19026 VC Base Board, there are five 2.54 mm pitch headers for IO extension. I2C and SWD pins from the WisDuo module are exposed on these headers.

J2 Header Pinout (I2C OLED display)

Pin	Pin Name	Description
1	3V3	3.3 V
2	GND	Ground pin
3	SCL	I2C1 clock

Pin	Pin Name	Description
4	SDA	I2C2 data

J3 Header Pinout

Pin	Pin Name	Description
1	3V3_S	controlled 3.3 V
2	GND	Ground pin
3	3V2	3.3 V
4	GND	Ground pin

J4 Header Pinout

Pin	Pin Name	Description
1	IO3	General purpose IO
2	IO4	General purpose IO
3	IO6	General purpose IO
4	IO7	General purpose IO

J7 Header Pinout

Pin	Pin Name	Description
1	SWDIO	Debug IO
2	SWDCLK	Debug CLK
3	GND	Ground pin
4	RESET	MCU Reset pin

J7 DEBUG HEADER

This header includes all required signals to connect a JLink or DAPLink debug/flash adapter.

J9 Header Pinout

Pin	Pin Name	Description
1	SCL	I2C1 clock
2	SDA	I2C2 data
3	AIN1	Analog input
4	IO8	General purpose IO

Battery and Solar Panel / 5V Connection

RAK19026 VC can be powered via the USB cable or Li-Ion/LiPo battery via the dedicated connectors, as shown in **Figure 7**. The matching connector for the battery wires is a depending on the board variant

- For EU/UK (CE & UKCA certification) JST PHR-3 2 mm pitch female .
- For other markets JST PHR-2 2 mm pitch female .

The battery can be recharged as well via a small solar panel or a regulated 5 V supply, as shown in **Figure 7**. The matching connector for the battery wires is a JST ZHR-2 1.5 mm pitch female .

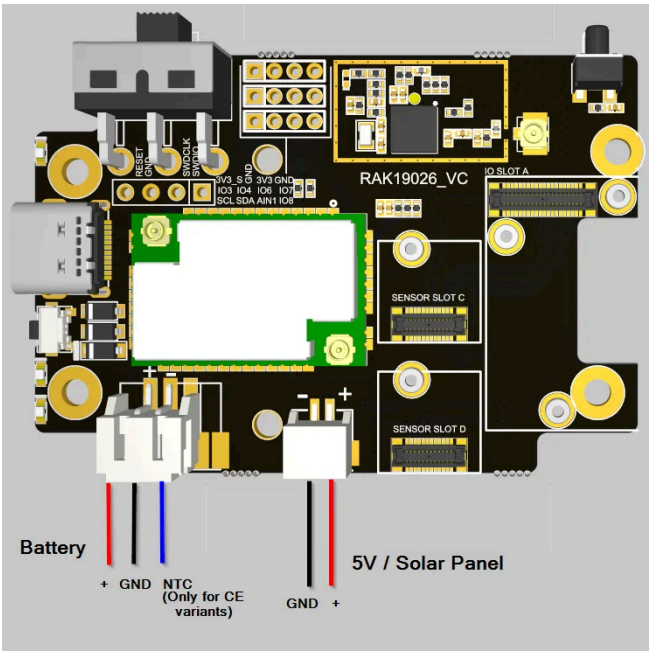


Figure 7: Battery connector pin order

IMPORTANT

The version of the RAK19026 with a 3-pin battery connector requires a matching battery with NTC temperature sensor and a 3-wire cable. If a different battery is used, the battery will not be charged.

WARNING

- Battery can cause harm if not handled properly.
- Only 3.7-4.2 V Rechargeable LiPo batteries are supported. It is highly recommended not to use other types of batteries with the system unless you know what you are doing.
- If a non-rechargeable battery is used, it has to be unplugged first before connecting the USB cable to the USB port of the board to configure the device. Not doing so might damage the battery or cause a fire.
- Make sure the battery wires match the polarity on the RAK19007 board. Not all batteries have the same wiring.

WARNING

- Only 5 V solar panels are supported. Do not use 12 V solar panels. It will destroy the charging unit and eventually other electronic parts.
- The GND pin of the Solar Panel Connector is located on edge of the board. Make sure the Solar Panel wires are matching the polarity on the RAK19007 board.

LEDs

Three LEDs are used to indicate the operating status. Below are the functions of the LEDs:

- 🔴 **Red LED** - Connected to the charger chip to indicate the charger status. When the battery is charging, this red LED is on. When the battery is full, this LED is weak light or off.
- 🟢 **Green LED** - Connected to the WisBlock Core module, controlled by MCU defined by the user.
- 🔵 **Blue LED** - Connected to the Wisblock Core module, controlled by MCU defined by the user.

RESET Push Button

The Reset Push Button is connected to the WisBlock Core module. When pushed, it resets the MCU.

GNSS Module

The GNSS module on the RAK19026 VC uses the u-blox ZOE-M8Q module. It supports a wide variety of satellite data protocols such as GPS, GLONASS, QZSS, and BeiDou. This ensures the retrieval of precise location data. The module features exceptional performance, high sensitivity, and minimal acquisition time — a very suitable module for your low-power IoT solution needs.

Features

- **Module Specification**
  - Uses the very accurate GNSS Module: **u-blox ZOE-M8Q chip**
  - Location Accuracy of ±2.5 meter
  - Velocity Accuracy of ±0.05 m/s
  - GPS, GLONASS, QZSS, and BeiDou Satellite support
  - Serial and I2C communication to WisBlock Core support
  - 10 Hz Update Rate
  - 29 seconds Location Fix from Cold Start, 1 second from Hot Start
  - Operating Voltage: 3.3 V
  - Operating Current: < 15 µA
  - Chipset: u-blox ZOE-M8Q

Chipset

The RAK19026 VC utilizes a very accurate u-blox ZOE-M8Q chip. See the manufacturer's u-blox ZOE-M8Q Page for more details.

Vendor	Part Number
u-blox	ZOE-M8Q

Acceleration Sensor

The acceration sensor on the RAK19026 VC is an ST LIS3DH 3-axis acceleration sensor. It comes with a ready-to-use SW library and tutorial, making it easy to build up a motion detection and acceleration data acquisition system. The sensor features an ultra-low-power high-performance three-axis linear accelerometer with a digital I2C interface. Additionally, it has ultra-low-power operational modes that allow advanced power saving and smart embedded functions.

The accelerometer of the RAK1904 module can be dynamically configured to work in the scales of ±2 g/±4 g/±8 g/±16g and is capable of measuring accelerations with output data rates from 1 Hz to 5.3 kHz.

Features

- **User selectable scales**: ±2g/±4g/±8g/±16g
- **Data acquisition rates**: from 1 Hz to 5.3 kHz
- **Voltage Supply**: 3.3 V
- **Current Consumption**: 0.5 uA to 11 uA
- **Chipset**: ST LIS3DH
- **Module size**: 10 x 10 mm#### Pin Definition

Chipset

Vendor	Part number
ST	LIS3DH

Connectors for WisBlock Sensor

The WisBlock sensor module connector is a **24-pin board-to-board connector**.

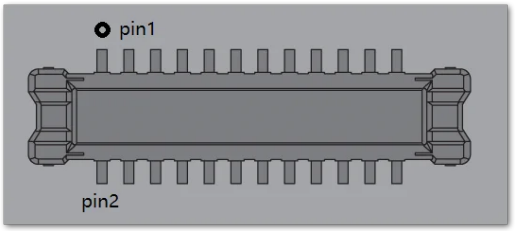


Figure 8: WisBlock Sensor module connector

> **NOTE**
>
> There are two connectors reserved for the sensor modules on the RAK19026 VC.

Connector D	Connector C	Pin Number	Pin Number	Connector C	Connector D
TXD1	NC	1	2	GND	GND
SPI_CS	SPI_CS	3	4	SPI_CLK	SPI_CLK
SPI_MISO	SPI_MISO	5	6	SPI_MOSI	SPI_MOSI
I2C1_SCL	I2C1_SCL	7	8	I2C1_SDA	I2C1_SDA
VDD	VDD	9	10	IO4	IO6
3V3_S	3V3_S	11	12	IO3	IO5
NC	NC	13	14	3V3_S	3V3_S
NC	NC	15	16	VDD	VDD
NC	NC	17	18	NC	NC
NC	NC	19	20	NC	NC
NC	NC	21	22	NC	NC
GND	GND	23	24	NC	RXD1

As for the following table, it shows the pin name and description of each pin in the WisBlock Sensor module connector.

Pin Number	Connector B	Connector C	Connector D	Type	Description
1	NC	NC	TXD1	I/O	UART TX signal
2	GND	GND	GND	S	Ground
3	SPI_CS	SPI_CS	SPI_CS	I/O	SPI chip select signal

Pin Number	Connector B	Connector C	Connector D	Type	Description
4	SPI_CLK	SPI_CLK	SPI_CLK	I/O	SPI clock
5	SPI_MISO	SPI_MISO	SPI_MISO	I/O	SPI MISO signal
6	SPI_MOSI	SPI_MOSI	SPI_MOSI	I/O	SPI MOSI signal
7	I2C1_SCL	I2C1_SCL	I2C1_SCL	I/O	I2C clock signal
8	I2C1_SDA	I2C1_SDA	I2C1_SDA	I/O	I2C data signal
9	VDD	VDD	VDD	S	Generated by CPU module. Used to power sensor board if MCU IO level is not 3.3 V
10	IO1	IO4	IO6	I/O	General purpose IO. IO2 controls the power switch of 3V3_S. When the 3V3_S function is used, IO2 can not be used as an interrupt of the sensor.
11	3V3_S	3V3_S	3V3_S	S	3.3 V power supply. Can be shut down by the CPU module.
12	IO2	IO3	IO5	I/O	General purpose IO - IO controls the power switch of 3V3_S. When the 3V3_S function is used, IO2 cannot be used as an interrupt of the sensor.
13	NC	NC	NC	NC	Not connected
14	3V3_S	3V3_S	3V3_S	S	3.3 V power supply. Can be shut down by the CPU module.
15	NC	NC	NC	NC	Not connected
16	VDD	VDD	VDD	S	Generated by CPU module. Used to power sensor board if the MCU IO level is not 3.3 V.
17	NC	NC	NC	NC	Not connected
18	NC	NC	NC	NC	Not connected
19	NC	NC	NC	NC	Not connected
20	NC	NC	NC	NC	Not connected
21	NC	NC	NC	NC	Not connected
22	NC	NC	NC	NC	Not connected
23	GND	GND	GND	S	Ground
24	NC	NC	RXD1	I/O	UART RX signal

Connector for WisBlock IO Slot

The WisBlock Module IO Slot connector, as shown in **Figure 9**, is a 40-pin board-to-board connector.

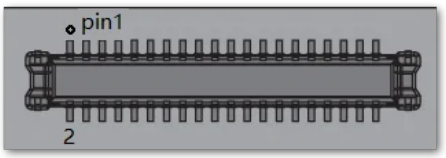


Figure 9: WisBlock IO slot connector

Pinout definition for IO slot:

Connector B	Connector A	Pin Number	Pin Number	Connector A	Connector B
VBAT	VBAT	1	2	VBAT	VBAT
GND	GND	3	4	GND	GND
3V3	3V3	5	6	3V3_S	3V3_S
USB+	USB+	7	8	USB–	USB–
VBUS	VBUS	9	10	SW1	SW1
TXD0	TXD0	11	12	RXD0	RXD0
RESET	RESET	13	14	LED1	LED1
LED2	LED2	15	16	LED3	LED3
VDD	VDD	17	18	VDD	VDD
I2C1_SDA	I2C1_SDA	19	20	I2C1_SCL	I2C1_SCL
AIN0	AIN0	21	22	AIN1	AIN1
NC	NC	23	24	NC	NC
SPI_CS	SPI_CS	25	26	SPI_CLK	SPI_CLK
SPI_MISO	SPI_MISO	27	28	SPI_MOSI	SPI_MOSI
IO1	IO1	29	30	IO2	IO2
IO3	IO3	31	32	IO4	IO4
TXD1	TXD1	33	34	RXD1	RXD1
I2C2_SDA	I2C2_SDA	35	36	I2C2_SCL	I2C2_SCL
IO5	IO5	37	38	IO6	IO6
GND	GND	39	40	GND	GND

As for the following table, it shows the pin name and description of the WisBlock IO module connector.

Pin Number	Pin Name	Type	Description
1	VBAT	S	Power supply from battery
2	VBAT	S	Power supply from battery
3	GND	S	Ground
4	GND	S	Ground
5	3V3	S	3.3 V power supply
6	3V3_S	S	3.3 V power supply. Can be shut down by a CPU module.

Pin Number	Pin Name	Type	Description
7	USB+	I/O	USB D+
8	USB–	I/O	USB D–
9	VBUS	S	5 V input for USB
10	SW1	I/O	User Defined Button (available on RAK4631/RAK4631-R and 11200 WisBlock Cores)
11	TXD0	I/O	MCU UART0 TX signal
12	RXD0	I/O	MCU UART0 RX signal
13	RESET	I	Connected to the reset switch, for MCU reset
14	LED1	I/O	LED for battery charge indicator
15	LED2	I/O	LED for custom used
16	LED3	I/O	LED for custom used
17	VDD	S	Generated by CPU module - Used for power sensor board if the MCU IO level is not 3.3 V
18	VDD	S	Generated by CPU module - Used for power sensor board if the MCU IO level is not 3.3 V.
19	I2C1_SDA	I/O	The first set of I2C data signal
20	I2C1_SCL	I/O	The first set of I2C clock signal
21	AIN0	A	Analog input for ADC
22	AIN1	A	Analog input for ADC
23	NC	NC	Not connect
24	NC	NC	Not connect
25	SPI_CS	I/O	SPI chip select signal
26	SP_CLK	I/O	SPI clock
27	SPI_MISO	I/O	SPI MISO signal
28	SPI_MOSI	I/O	SPI MOSI signal
29	IO1	I/O	General purpose IO
30	IO2	I/O	Used for 3V3_S enable
31	IO3	I/O	General purpose IO
32	IO4	I/O	General purpose IO
33	TXD1	I/O	MCU UART1 TX signal
34	RXD1	I/O	MCU UART1 RX signal
35	I2C2_SDA	I/O	The second set of I2C data signal

Pin Number	Pin Name	Type	Description
36	I2C2_SCL	I/O	The second set of I2C clock signal
37	IO5	I/O	General purpose IO
38	IO6	I/O	General purpose IO
39	GND	S	Ground
40	GND	S	Ground

Electrical Characteristics

Absolute Maximum Ratings

The Absolute Maximum Ratings of the device are shown in the table below. The stress ratings are the functional operation of the device.

> **WARNING**
>
> 1. If the stress rating goes above what is listed, it may cause permanent damage to the device.
> 2. Under the listed conditions is not advised.
> 3. Exposure to maximum rating conditions may affect the device reliability.

Ratings	Maximum Value	Unit
Power supply on the USB port (**VBUS**)	–0.3 to 5.5	V
Battery voltage (**VBAT**)	–0.3 to 4.3	V
Solar panel voltage (**CONN_S**)	–0.3 to 5.5	V
IOs of WisBlock connector	–0.3 to VDD+0.3	V
ESD	2000	V

> **WARNING**
>
> The RAK19026 VC, as any electronic equipment, is sensitive to **electrostatic discharge (ESD)**. Improper handling can cause permanent damage to the module.

Current Consumption

The RAK19026 VC is designed for **low-power IoT products**, and the power supply uses a high-efficiency low grounding current regulator. When there is no module on RAK19026 VC, the **leakage current is lower than 2 µA**. With WisBlock Core and WisBlock Sensor on it, the sleep current is **lower than 10 µA**. When a LoRa module is transmitting, the current may reach **130 mA**.

Conditions	Current	Unit
Leakage current, without any module on RAK19026 VC	2	µA
Idle current, with MCU and sensors in sleep mode	10	µA
Working current, with LoRa module transmitting	130	mA

Battery Connector

The RAK19026 VC WisMesh Base can be powered by a rechargeable battery, connected to the **P2 connector**. The nominal operating voltage of the battery should be within the range shown in the following table. The matching connector for the battery wires is an JST PHR-2 2 mm pitch female .

Minimum	Typical	Maximum	Unit
3.3	3.7	4.3	V

Pinout

Pin	Function	Remark
1	+	3.7-4.2 V
2	GND	GND
3	NTC	Only for EU/UK models

The Type-C USB connector is used as a charging port. The voltage and current fed to the battery through the port should not exceed its charging limits, as shown in the table below.

Parameter	Value
Charging voltage	4.5 – 5.5 V
Charging current	350 mA

Pinout

Pin	Function	Remark
1	+	4.5 – 5.5 V
2	GND	GND

A suitable Li-Ion battery should have the following parameters as shown in the table below:

Parameter	Value
Standard voltage	3.7 V
Charging voltage	4.2 V
Capacity	As required
Discharge current	At least 500 mA

IMPORTANT

For EU/UK market, the battery ***must*** have an NTC temperature sensor with 10 kΩ nominal resistance and a 3 wire battery cable matching with the pin out of the battery connector

WARNING

Do not use a non-rechargeable battery.

Solar Panel Connector

A 5 V solar panel can be connected to the board via the **P1 connector**. The solar panel can also be used to charge the Li-Ion battery. The matching connector for the solar panel wires is an JST ZHR-2 1.5 mm pitch female .

Mechanical Characteristics

Board Dimensions

> **NOTE**
>
> - You may also refer and download the M1.2 Stand-off fastener/inserts datasheet .

Top components

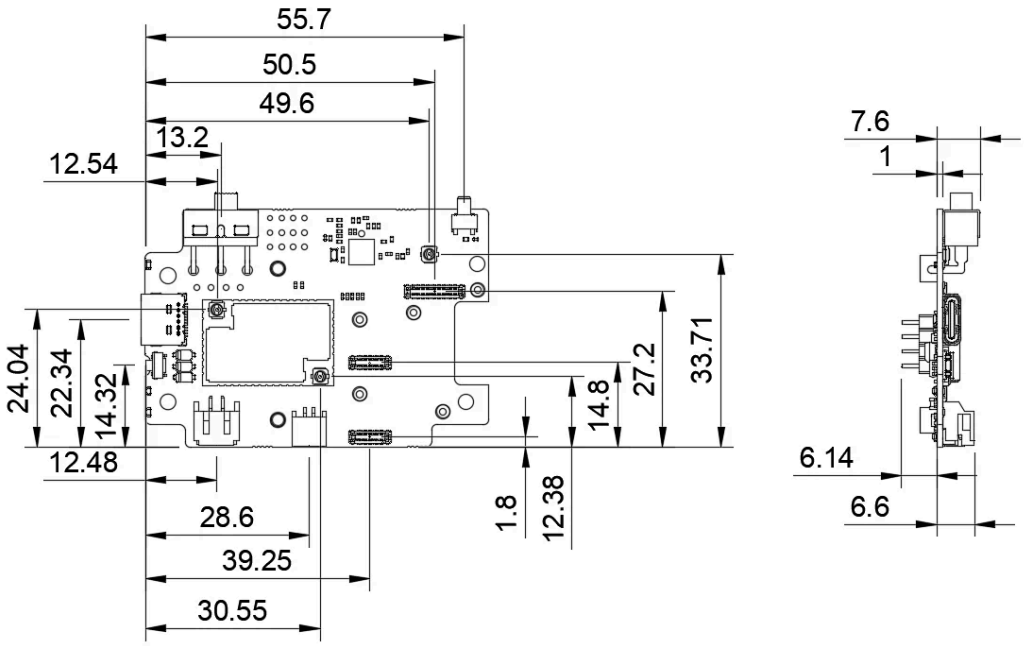


Figure 10: RAK19026 VC mechanical dimensions

Bottom components

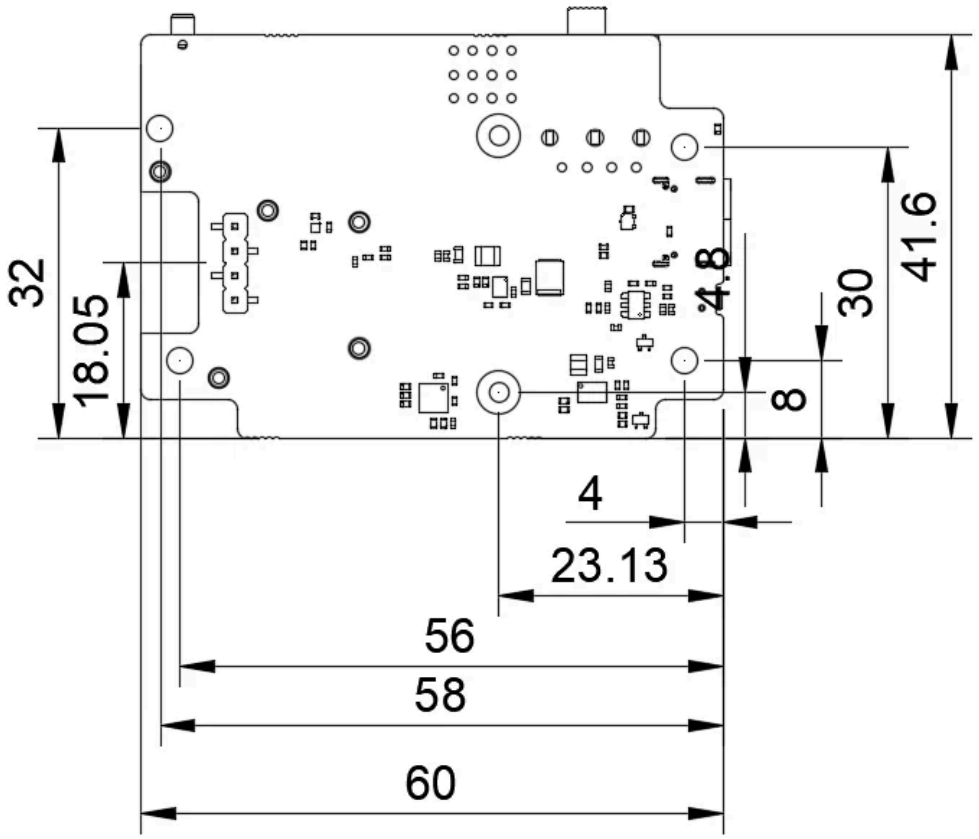


Figure 11: RAK19026 VC mounting holes location and diameter top view

WisConnector PCB Layout

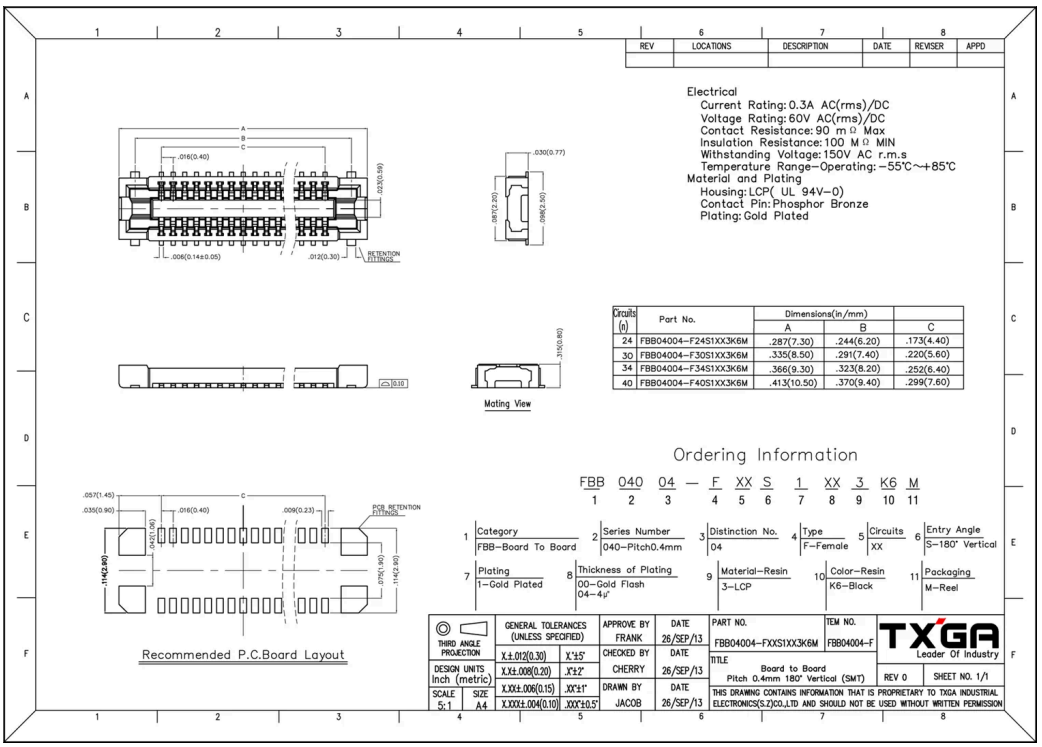


Figure 12: WisConnector PCB footprint and recommendations

Environmental Characteristics

The table below lists the operation and storage temperature requirements of RAK19026 VC:

Parameter	Minimum	Typical	Maximum
Operational Temperature Range	–35° C	+25° C	+75° C
Extended Temperature Range	–40° C	+25° C	+80° C
Storage Temperature Range	–40° C	+25° C	+80° C

Schematic Diagram

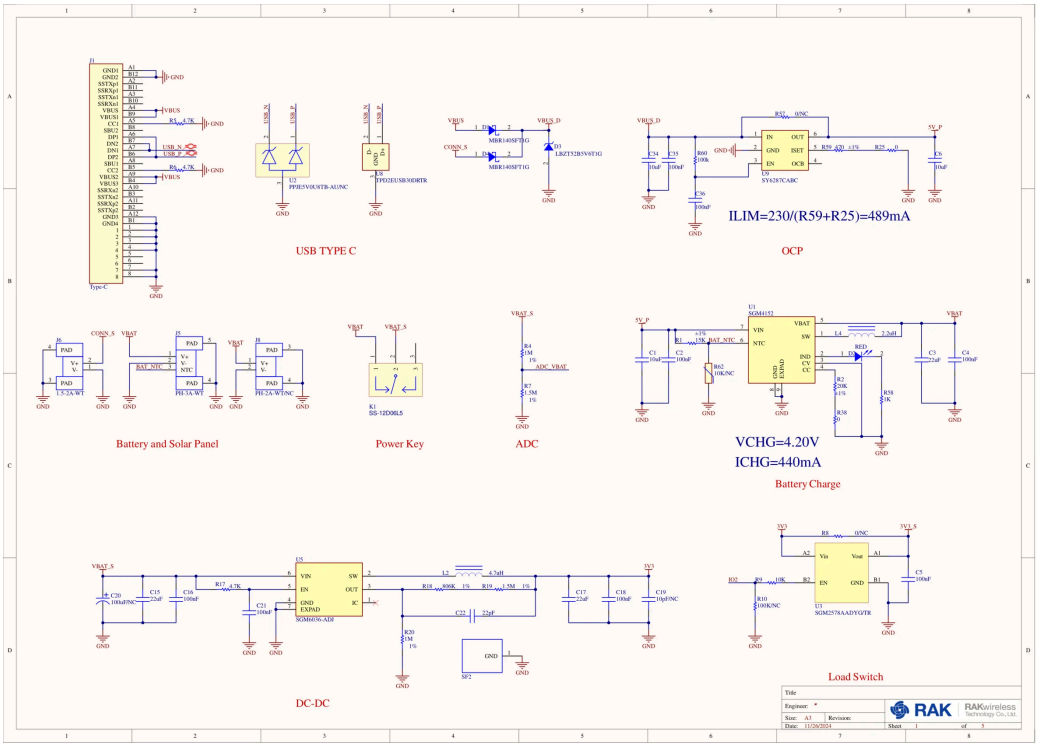


Figure 13: RAK19026 VC schematic diagram (Power)

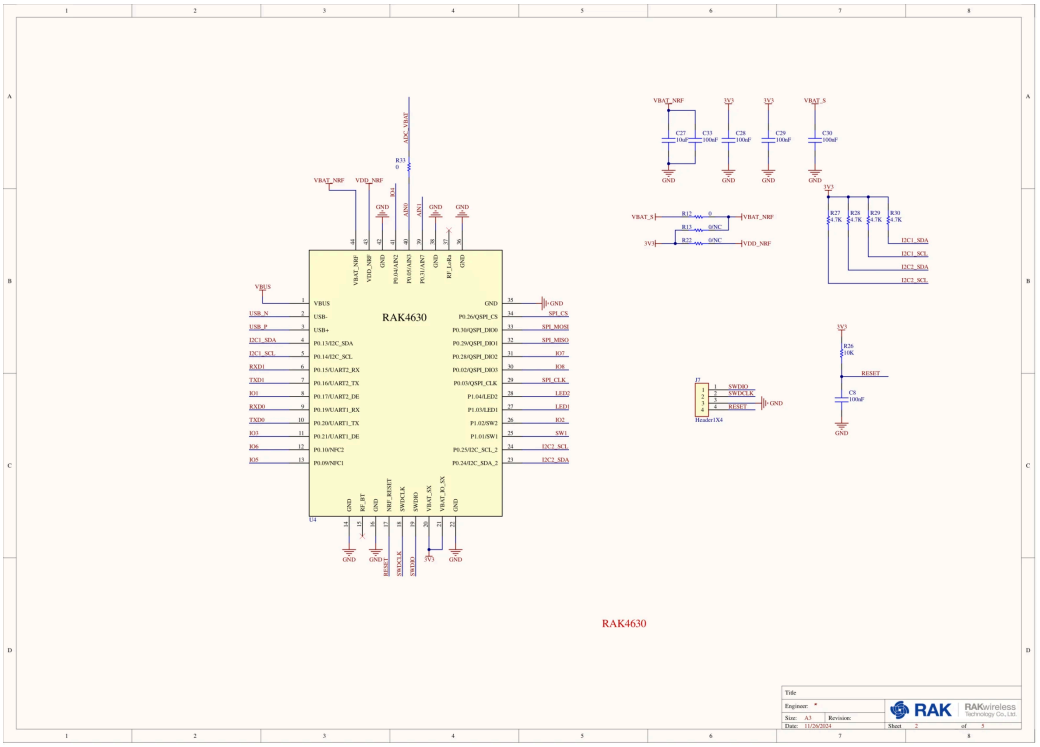


Figure 14: RAK19026 VC schematic diagram (RAK4630)

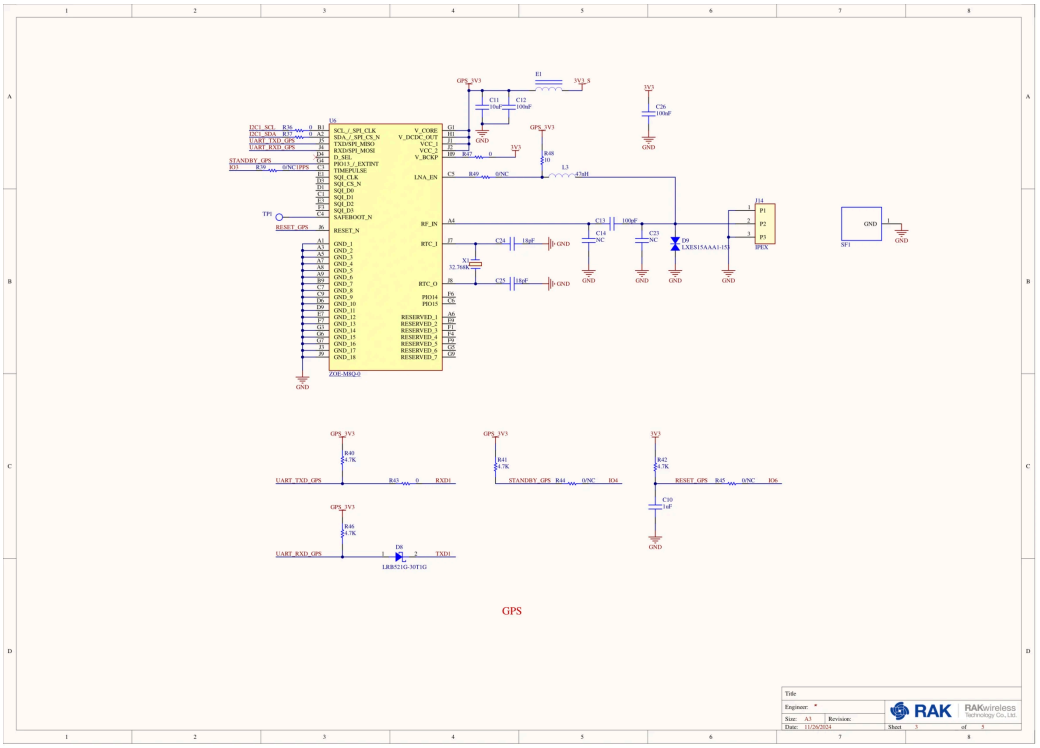


Figure 15: RAK19026 VC schematic diagram (GNSS)

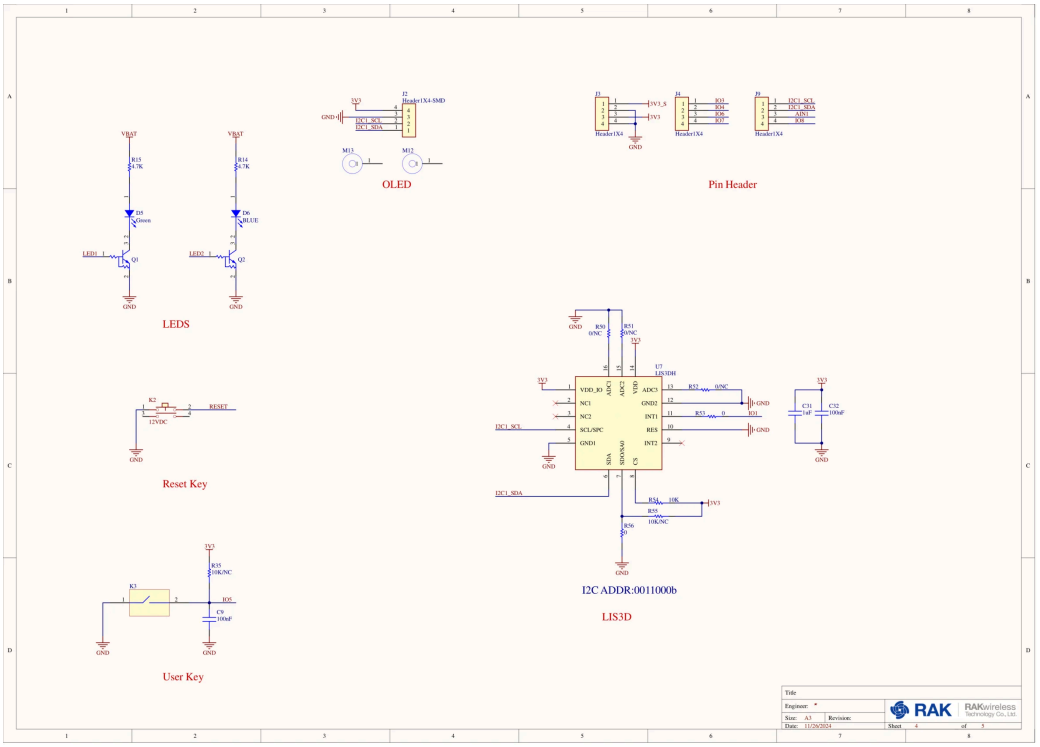


Figure 16: RAK19026 VC schematic diagram (Pin Headers - Acceleration Sensor)

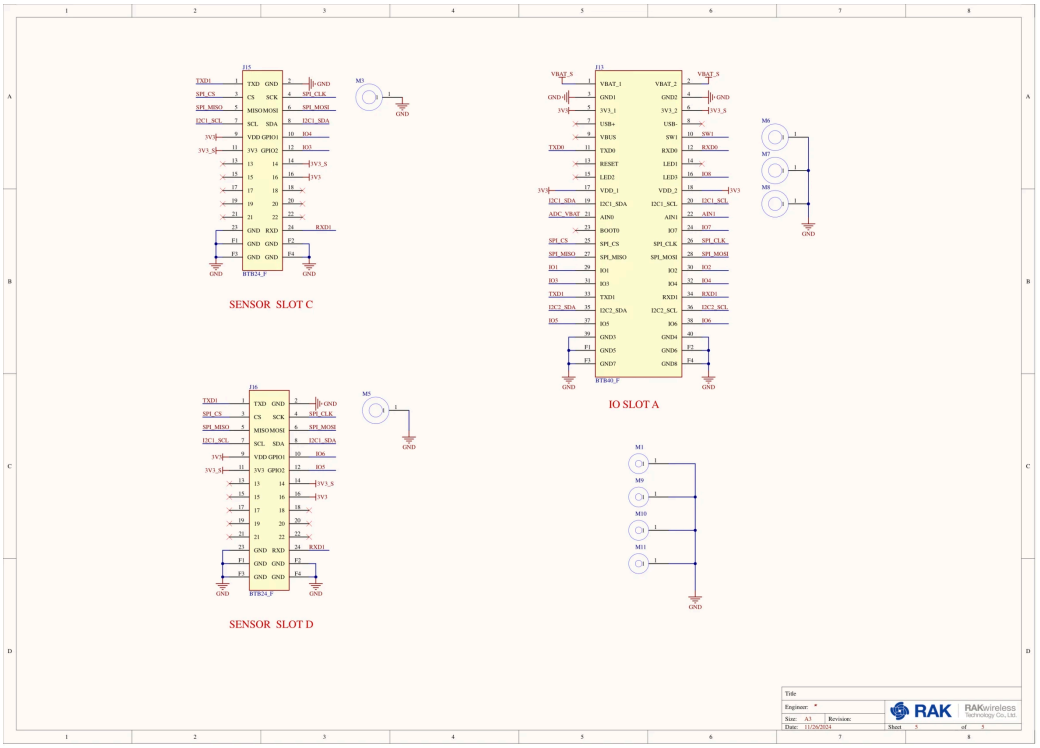


Figure 17: RAK19026 VC schematic diagram (Module Slots)